MRG-ViT–Mamba Architecture Presentation

Reliability-Guided ViT + Mamba Temporal Modeling for Student Engagement Recognition

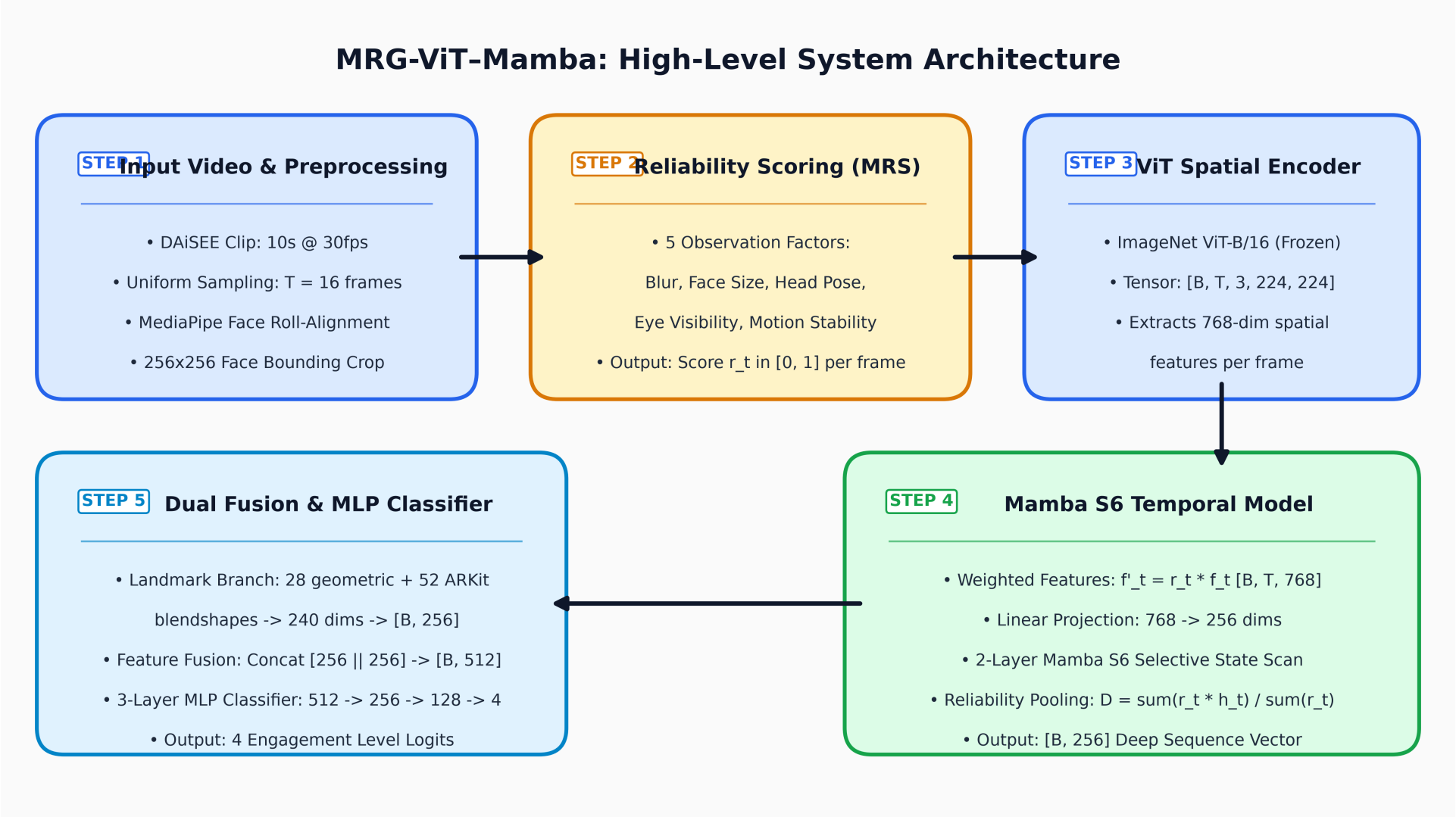
DAiSEE Benchmark Summary & Pipeline Verification Report

Executive Summary: This presentation details the architecture, verification, and evaluation of the MRG-ViT–Mamba model. The system combines a frozen Vision Transformer (ViT-B/16) spatial encoder, a Motion Reliability Score (MRS) quality filter, a Mamba selective state-space temporal model (S6), and MediaPipe facial blendshapes for 4-class student engagement classification.

Parameter / Setting	Specification Detail
Spatial Encoder	ImageNet ViT-B/16 (Frozen 85.8M parameters)
Temporal SSM	2-Layer Mamba S6 (d_model=256, d_state=16, d_conv=4)
Landmark Features	28 Geometric Metrics + 52 ARKit Blendshapes (L=80 per frame)
Total Trainable Parameters	1,365,892 (~1.37 Million parameters)
Pipeline Verification	100% Passed (38/38 Unit Tests Passed Cleanly)

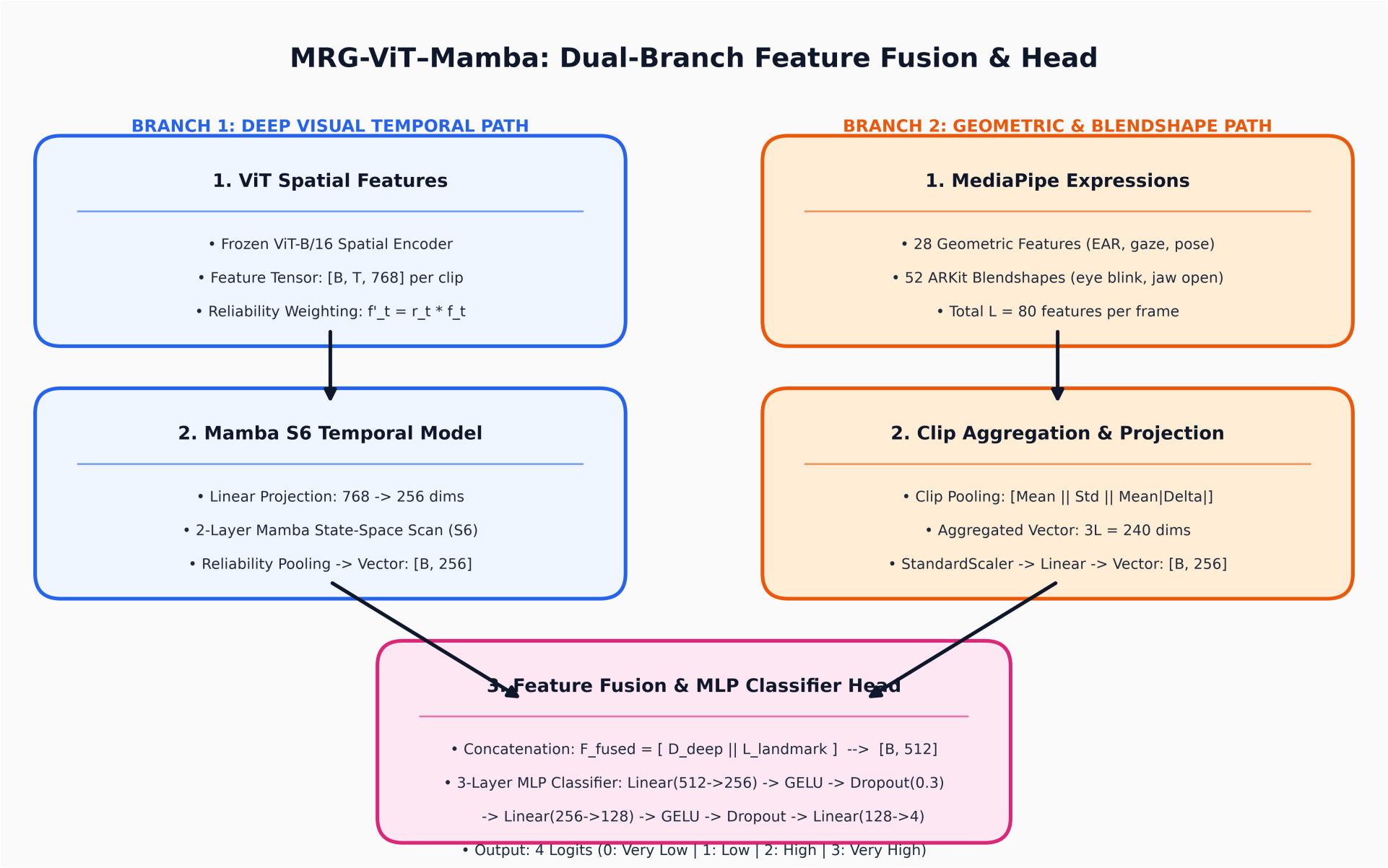
1. High-Level System Flow Architecture

Overview of the 5 main stages from raw DAiSEE video to 4-class engagement predictions:



2. Dual-Branch Feature Fusion Architecture

Detailed breakdown of Deep ViT-Mamba Branch vs MediaPipe Blendshape Branch:



3. Motion Reliability Score & Verification Results

Motion Reliability Score (MRS) Formula:
MRS_t = 0.2·Blur + 0.2·FaceSize + 0.2·Pose + 0.2·Eyes + 0.2·Motion
MRS quantifies observation quality per frame in range [0.0, 1.0] without touching engagement labels.

Verification Checkpoint Summary (38/38 Unit Tests Passed):

Checkpoint / Test	Status	Verified Evidence
Unit Test Suite (tests/test_units.py)	**PASS**	38/38 unit tests passed cleanly in 12.67s.
Dataset & Split Audit	**PASS**	108/108 decodable clips; 0 subject overlap.
Face & Landmark Pipeline	**PASS**	100% face detection on test frames, 99.48% corpus.
Motion Reliability Score (MRS)	**PASS**	MRS in [0.2976, 0.9880], std=0.1272. No NaNs.
ViT & Mamba S6 Causality	**PASS**	ViT [B,T,768]; Mamba S6 causality verified (0.0 at t<8).
Full Overfit Gate	**PASS**	Gradient updates on 35 tensors; tiny overfit loss -> 0.0000.